

SENSORS FOR T&M: Sensing The World

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A sensor is required in every stage of product development. This article looks at the various sensors available for the purpose of test and measurement

An electronic sensor detects and measures a physical phenomenon, such as pressure, force, temperature or acceleration, and provides a corresponding output in the form of an electronic signal like voltage, current or frequency. Depending on the application, different sensors are used for different parameters. For example, different types of pressure sensors are used to measure atmospheric pressure and pressure in the tyres of heavy vehicles.

The global marketplace for sensors is driven by innovation and development. A sensor is required in every stage of product development, including research and development (R&D), production variation control, and process monitoring and protection. Test and measurement (T&M) supports the sensing needs of those involved in development, product research, test programs, test and evaluation. Latest technologies enable the development of customised solutions for a wide range of applications. Some manufacturers also design sensors as per customer requirement and testing applications.

Typical sensors for T&M

Discussed here are some common sensors, their applications and requirements in different areas.

Pressure sensor. This type of sensor has wide applications for measuring pressure in commercial vehicles, passenger vehicles, tyre profiling, tyre contact analysis, tyre bead pressure, atmospheric pressure and heavy equipment used in agriculture, mining, forestry and construction. It minimises the effects of hysteresis and compression while designing.

To meet the most demanding T&M specifications, miniature pressure sensors are manufactured for rugged environments. Industrial processes need highly specialised pressure sensors to be used in hazardous locations.

Temperature sensor. This is the most frequently used sensor for measuring temperature in a cooling tower for thermal sensing, electric motor winding, bearings, engine oil, laboratories during different test procedures, solar applications, geo-thermal plants, rail vehicles, deep freezing technologies, health services and more.

Motion sensor. This sensor is primarily used for detecting moving objects (especially people) and gathering data regarding position, acceleration and velocity for safety and security applications.

Speed sensor. This detects speed of an object or vehicle. It is required for speedometers, lidars, ground speed radars, pitometer logs, doppler radars, air-speed indicators and so on.

Acceleration sensor. This sensor measures the gravitational force in the direction of the sensor. It is used in mobiles, robots and aerospace applications, among others, where measuring acceleration is a major component.

Position (lidar) sensor. It measures position that can be absolute or relative. This sensor finds application in aircraft, high-speed vehicles, factory automation and security applications. The position can be linear or angular. A magnetic sensor is used for digital and linear outputs of the angle.

Micro-electro-mechanical systems (MEMS) microphone sensor. This sensor targets all audio applications where key

Fig. 1: Setup for gas-sensing measurement
(Credit: Research Gate)

